

Level-Set Deficit Identity for the Nicola–Tilli Route

This note implements the first concrete step from `plan2.md`: turn the convexity gap in `faber-krahn-1.pdf` into an exact weighted integral of the two losses in the level-set proof.

Throughout, let $u = u_\Omega$ solve

$$-\Delta u = 1 \quad \text{in } \Omega, \quad u = 0 \quad \text{on } \partial\Omega,$$

and set

$$E_t = \{u > t\}, \quad m(t) = |E_t|, \quad \Sigma_t = \{u = t\}, \quad P(t) = \mathcal{H}^{n-1}(\Sigma_t).$$

The formulas below are written for smooth domains and regular levels. The same identities should extend to the finite-perimeter/coarea setting by the usual approximation and a.e. level-set interpretation.

1 The Two Pointwise Losses

For a regular level t , define

$$a(t) := -m'(t) = \int_{\Sigma_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

By the divergence theorem applied to E_t ,

$$\int_{\Sigma_t} |\nabla u| d\mathcal{H}^{n-1} = m(t).$$

Let

$$c_n := n^2 \omega_n^{2/n}.$$

The total level-set loss in the differential inequality is

$$D_{\text{lev}}(t) := a(t)m(t) - c_n m(t)^{2-2/n}.$$

It splits as

$$D_{\text{lev}}(t) = D_H(t) + D_I(t),$$

where

$$D_H(t) := \left(\int_{\Sigma_t} \frac{1}{|\nabla u|} \right) \left(\int_{\Sigma_t} |\nabla u| \right) - P(t)^2 = a(t)m(t) - P(t)^2$$

is the Holder/Cauchy–Schwarz loss, and

$$D_I(t) := P(t)^2 - c_n m(t)^{2-2/n}$$

is the squared-perimeter isoperimetric loss of E_t .

Both terms are nonnegative: $D_H(t) \geq 0$ by Cauchy–Schwarz, and $D_I(t) \geq 0$ by the isoperimetric inequality.

2 Convexity Variable and Exact Identity

Let $L := |\Omega|$, let $v(s) = u^*(s) = m^{-1}(s)$, and define

$$I(s) := \int_{\{u > v(s)\}} u \, dx.$$

Equivalently, for regular values,

$$I'(s) = v(s), \quad I''(s) = v'(s) = \frac{1}{m'(v(s))} = -\frac{1}{a(v(s))}.$$

The function used in `faber-krahn-1.pdf` is

$$G(s) := I(s) + \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}}.$$

Since

$$\frac{d^2}{ds^2} \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{1}{c_n s^{1-2/n}},$$

we get, with $s = m(t)$,

$$G''(s) = \frac{1}{c_n s^{1-2/n}} - \frac{1}{a(t)} = \frac{D_{\text{lev}}(t)}{s a(t) c_n s^{1-2/n}}.$$

The endpoint convexity gap is

$$\Gamma(\Omega) := G'(L) - \frac{G(L)}{L}.$$

Because $G(0) = 0$,

$$\Gamma(\Omega) = \frac{1}{L} \int_0^L s G''(s) \, ds.$$

Changing variables $s = m(t)$, so that $ds = -a(t) \, dt$, gives the exact deficit identity

$$\Gamma(\Omega) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} \, dt.$$

This is the clean quantitative form of the rigidity statement at the end of `faber-krahn-1.pdf`.

3 Relation to the Saint-Venant Deficit

At $s = L$, one has $v(L) = 0$. Hence

$$G'(L) = \frac{L^{2/n}}{2n\omega_n^{2/n}}.$$

Also

$$\frac{G(L)}{L} = \frac{1}{L} \int_{\Omega} u \, dx + \frac{L^{2/n}}{(4+2n)\omega_n^{2/n}}.$$

Therefore

$$\Gamma(\Omega) = \frac{1}{L} \left(\int_B u_B \, dx - \int_{\Omega} u_{\Omega} \, dx \right),$$

where B is the ball with $|B| = |\Omega|$. Since

$$E(\Omega) = -\frac{1}{2} \int_{\Omega} u_{\Omega} dx,$$

we obtain

$$\Gamma(\Omega) = \frac{2}{|\Omega|} (E(\Omega) - E(B)).$$

Thus the endpoint convexity gap is not merely comparable to the torsion deficit; it is exactly the normalized Saint-Venant deficit.

4 Consequences for Good Levels

The quantitative isoperimetric inequality gives, for a dimensional constant $c > 0$,

$$D_I(t) \geq c m(t)^{2-2/n} \mathcal{A}(E_t)^2$$

for a.e. regular level t . Hence

$$E(\Omega) - E(B) \gtrsim_n \int_0^{\|u\|_{\infty}} m(t) \mathcal{A}(E_t)^2 dt$$

up to the normalization factor $|\Omega|$. More precisely, the exact identity controls the weighted average

$$\int_0^{\|u\|_{\infty}} \frac{D_I(t)}{m(t)^{1-2/n}} dt.$$

For any measurable level interval $J \subset (0, \|u\|_{\infty})$, define the weight

$$d\nu(t) := \frac{dt}{n^2 \omega_n^{2/n} m(t)^{1-2/n}}.$$

If $\nu(J) > 0$, then there exists a regular $t \in J$ such that

$$D_H(t) + D_I(t) \leq \frac{|\Omega| \Gamma(\Omega)}{\nu(J)}.$$

In particular, on any slab where $m(t)$ is bounded above and below, the Saint-Venant deficit produces at least one level with both small isoperimetric loss and small Holder loss.

5 Holder Loss as a Serrin-Variance Defect

Let $f = |\nabla u|$ on Σ_t , and set

$$\bar{f} := \frac{1}{P(t)} \int_{\Sigma_t} f = \frac{m(t)}{P(t)}.$$

Then the Holder deficit has the exact weighted variance representation

$$\int_{\Sigma_t} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} = \frac{m(t)}{P(t)^2} D_H(t).$$

Indeed,

$$\int_{\Sigma_t} \frac{(f - \bar{f})^2}{f} = \int_{\Sigma_t} f - 2\bar{f}P(t) + \bar{f}^2 \int_{\Sigma_t} \frac{1}{f} = \frac{m(t)}{P(t)^2} \left(m(t) \int_{\Sigma_t} \frac{1}{f} - P(t)^2 \right).$$

Thus small $D_H(t)$ says that $|\nabla u|$ is almost constant on Σ_t , in the natural weighted $L^2(1/f)$ sense.

If, on a good regular level, one also has

$$0 < c_t \leq |\nabla u| \leq C_t < \infty,$$

then this becomes an ordinary L^2 -variance estimate:

$$\int_{\Sigma_t} |f - \bar{f}|^2 d\mathcal{H}^{n-1} \leq C_t \frac{m(t)}{P(t)^2} D_H(t).$$

This is the precise bridge from the Nicola–Tilli convexity gap to an approximate Serrin condition on the interior domain E_t .

6 What Remains Open in This Route

The identity proves that small Saint-Venant deficit forces many superlevel sets to be simultaneously almost isoperimetric and almost overdetermined.

The unsolved propagation problem is to recover $\mathcal{A}(\Omega)^2$ from this information. The deficit identity sees interior levels with positive height, while Fraenkel asymmetry is a $t = 0$ boundary datum. A sharp proof by this route needs a boundary-layer theorem of the following form:

$$\exists t \downarrow 0 \quad \text{such that} \quad |\Omega \setminus E_t| \text{ is small and } D_H(t) + D_I(t) \text{ remains controlled.}$$

Without such a theorem, the level-set method gives strong information on interior cores E_t , but not yet a sharp estimate for the original domain Ω .